

Problem Set 3: Answers

Probability and Statistical Inference

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Reference formulas for Question 1. As in Lecture 3, choosing a model matters; memorising the gallery formulas is non-examinable. You may use the following.

Distribution	Mean; variance	Useful probability
$N \sim \text{Poisson}(\lambda)$	$\lambda; \lambda$	$\Pr(N = j) = e^{-\lambda} \lambda^j / j!$
$W \sim \text{Exponential}(\lambda)$	$1/\lambda; 1/\lambda^2$	$\Pr(W > w) = e^{-\lambda w}, w \geq 0$
$G \sim \text{Gamma}(a, \lambda)$	$a/\lambda; a/\lambda^2$	shape–rate
$B \sim \text{Binomial}(m, p)$	$mp; mp(1-p)$	
$H \sim \text{Geometric}(p)$	$1/p; (1-p)/p^2$	$H = 1, 2, \dots$

1. **Choosing probability distributions.** Use the model for counts and waiting times from Lecture 3 for calls arriving at a support center at a constant rate of 3 per hour. Counts in disjoint time intervals are independent. Independently across calls, each call results in a sale with probability 0.2.
 - (a) Let N be the number of calls in one hour. State its distribution, find $\mathbb{P}(N = 2)$, and give its expectation and variance.
 - (b) Let W_1 be the waiting time, in hours, to the next call. State its distribution, find $\mathbb{P}(W_1 > 1/2)$, and give its expectation.
 - (c) Let W_4 be the waiting time to the fourth call. State its distribution using the shape–rate parameterization, and give its expectation and variance.
 - (d) Let S be the number of sales among the next ten calls, and let H be the number of calls up to and including the first sale. State both distributions. Find $\mathbb{E}[S]$, $\text{Var}(S)$, $\mathbb{P}(H > 4)$ and $\mathbb{E}[H]$.
 - (e) A researcher needs a distribution for an unknown proportion that can take values only between zero and one. The researcher also needs a distribution for a positive variable produced by repeated multiplicative changes (for example, repeated percentage growth). Which two distributions from the lecture are natural starting points? Briefly explain each choice.

Solution.

- (a) The number of arrivals in a fixed one-hour window is

$$N \sim \text{Poisson}(3).$$

Consequently,

$$\mathbb{P}(N = 2) = e^{-3} \frac{3^2}{2!} = \frac{9}{2} e^{-3} \simeq 0.2240, \quad \mathbb{E}[N] = \text{Var}(N) = 3.$$

- (b) The waiting time until the next Poisson arrival has an exponential distribution with the same rate:

$$W_1 \sim \text{Exponential}(3).$$

The probability that the wait exceeds w hours is $\mathbb{P}(W_1 > w) = e^{-3w}$, so

$$\mathbb{P}(W_1 > 1/2) = e^{-3/2} \simeq 0.2231, \quad \mathbb{E}[W_1] = \frac{1}{3} \text{ hour.}$$

- (c) The waiting time to the fourth arrival is

$$W_4 \sim \text{Gamma}(4, 3),$$

where the first parameter is shape and the second is rate. Thus

$$\mathbb{E}[W_4] = \frac{4}{3} \text{ hours}, \quad \text{Var}(W_4) = \frac{4}{3^2} = \frac{4}{9} \text{ hours}^2.$$

- (d) The ten sale indicators are independent Bernoulli variables with success probability 0.2, so

$$S \sim \text{Binomial}(10, 0.2), \quad \mathbb{E}[S] = 10(0.2) = 2, \quad \text{Var}(S) = 10(0.2)(0.8) = 1.6.$$

The number of calls through the first sale is

$$H \sim \text{Geometric}(0.2)$$

on $1, 2, \dots$. The event $H > 4$ means that the first four calls all fail to produce a sale. Hence

$$\mathbb{P}(H > 4) = 0.8^4 = 0.4096, \quad \mathbb{E}[H] = \frac{1}{0.2} = 5.$$

- (e) A beta distribution is a natural model for a proportion because it takes values between zero and one, and its two shape parameters allow many symmetric and asymmetric shapes. A log-normal distribution is a natural starting point for multiplicative growth. If repeated multiplicative changes make the log of the final variable approximately normal, then the variable itself is positive and log-normal, usually with right skew.

2. Normal vectors and related distributions. Suppose

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 4 & 2 \\ 2 & 9 \end{pmatrix}\right).$$

- (a) Give the marginal distributions of X_1 and X_2 , and calculate their correlation.
 (b) Find the distribution of $Y = 3 + X_1 - 2X_2$.
 (c) Find the value of a for which X_1 and $X_2 - aX_1$ are independent. Also give the distribution of $X_2 - aX_1$ at this value of a . Explain carefully which step of your argument uses joint normality.
 (d) Write $\mu = \mathbb{E}[X]$ and $\Sigma = \text{Var}(X)$. By multiplying out, verify that

$$(X - \mu)' \Sigma^{-1} (X - \mu) = \left(\frac{X_1 - 1}{2}\right)^2 + \left(\frac{X_2 - X_1/2 + 3/2}{\sqrt{8}}\right)^2.$$

Use part (c) and the lecture's definition of χ^2 to find its distribution.

- (e) Let $Z \sim N(0, 1)$ and $Q \sim \chi_\nu^2$ be independent, and define $T = Z/\sqrt{Q/\nu}$. Use the definitions of the chi-squared and F distributions to show that $T^2 \sim F_{1, \nu}$.

Solution.

- (a) Marginals of a multivariate normal vector are normal. Reading the means and diagonal variances from the parameter matrices gives

$$X_1 \sim N(1, 4), \quad X_2 \sim N(-1, 9).$$

Their covariance is 2, while their standard deviations are 2 and 3. Therefore

$$\text{Corr}(X_1, X_2) = \frac{2}{2 \cdot 3} = \frac{1}{3}.$$

- (b) A linear transformation of a normal vector is normal. Its mean is

$$\mathbb{E}[Y] = 3 + 1 - 2(-1) = 6.$$

Writing $b = (1, -2)'$, its variance is

$$\text{Var}(Y) = b' \text{Var}(X) b = 4 + (-2)^2(9) + 2(1)(-2)(2) = 32.$$

Thus $Y \sim N(6, 32)$.

- (c) We first choose a to make the covariance zero:

$$\text{Cov}(X_1, X_2 - aX_1) = \text{Cov}(X_1, X_2) - a \text{Var}(X_1) = 2 - 4a.$$

It vanishes when $a = 1/2$. At this value,

$$\begin{aligned} \mathbb{E}[X_2 - \tfrac{1}{2}X_1] &= -1 - \tfrac{1}{2} = -\tfrac{3}{2}, \\ \text{Var}(X_2 - \tfrac{1}{2}X_1) &= 9 + \tfrac{1}{4}(4) - 2(\tfrac{1}{2})(2) = 8. \end{aligned}$$

Hence $X_2 - X_1/2 \sim N(-3/2, 8)$. The transformed pair $(X_1, X_2 - X_1/2)$ is jointly normal because it is a linear transformation of X . Zero covariance implies independence for a jointly normal pair. Without joint normality, making the covariance zero would not by itself establish independence.

- (d) The determinant of Σ is 32, so

$$\Sigma^{-1} = \frac{1}{32} \begin{pmatrix} 9 & -2 \\ -2 & 4 \end{pmatrix}.$$

For $u = X_1 - 1$ and $v = X_2 + 1$, both sides of the proposed identity equal

$$\frac{9u^2 - 4uv + 4v^2}{32} = \frac{u^2}{4} + \frac{(v - u/2)^2}{8}.$$

By parts (a) and (c), $(X_1 - 1)/2$ and $(X_2 - X_1/2 + 3/2)/\sqrt{8}$ are independent standard normal variables. Their squared sum therefore has distribution χ_2^2 .

- (e) Squaring the definition of T gives

$$T^2 = \frac{Z^2}{Q/\nu} = \frac{Z^2/1}{Q/\nu}.$$

By the definition of a chi-squared variable, $Z^2 \sim \chi_1^2$. Moreover, Z^2 remains independent of Q because it is a function of the independent variable Z . The displayed ratio is therefore a ratio of two independent chi-squared variables, each divided by its degrees of freedom. This is exactly the definition of $F_{1,\nu}$, so $T^2 \sim F_{1,\nu}$.

3. **Tail behavior and moments.** For $a > 0$, consider the following candidate probability density. Calling it a candidate means that you must still check that it is a valid density:

$$f_a(x) = \frac{a}{2(1 + |x|)^{a+1}}, \quad -\infty < x < \infty.$$

- (a) Verify that f_a is a probability density.
- (b) For $r > 0$, determine exactly when $\mathbb{E}[|X|^r]$ is finite. It is sufficient to compare the integrand with a power of x in the tails; you do not need to evaluate the moment when it is finite.
- (c) For which values of a does the mean exist? When it exists, what is it? Explain why symmetry alone is not enough to answer this question.
- (d) For which values of a does the variance exist? Give one value of a for which the mean exists but the variance does not.
- (e) For $M \geq 0$, show that $\mathbb{P}(|X| > M) = (1 + M)^{-a}$. Explain why a tail probability can approach zero even when some moments are infinite, so this result does not contradict your earlier answers.

Solution.

- (a) The function is nonnegative. By symmetry and the substitution $u = 1 + x$,

$$\int_{-\infty}^{\infty} f_a(x) dx = a \int_0^{\infty} (1+x)^{-a-1} dx = a \left[\frac{(1+x)^{-a}}{-a} \right]_0^{\infty} = 1.$$

It is therefore a valid density.

- (b) Again using symmetry,

$$\mathbb{E}[|X|^r] = a \int_0^{\infty} \frac{x^r}{(1+x)^{a+1}} dx.$$

Near zero, the integrand is finite and causes no problem. For large x , it is comparable to

$$\frac{x^r}{x^{a+1}} = x^{r-a-1}.$$

The integral of this power over $[1, \infty)$ is finite exactly when $r - a - 1 < -1$, or $r < a$. At the boundary $r = a$, the integrand behaves like $1/x$ and the integral diverges. Thus

$$\mathbb{E}[|X|^r] < \infty \iff r < a.$$

- (c) A finite mean requires $\mathbb{E}|X| < \infty$, which by part (b) holds exactly when $a > 1$. In that case the odd integrand $xf_a(x)$ is absolutely integrable and symmetry gives $\mathbb{E}[X] = 0$. When $a \leq 1$, the positive-tail integral is $+\infty$ and the negative-tail integral is $-\infty$. Canceling these as though they were finite would create the undefined expression $\infty - \infty$. Symmetric truncation of the two tails would give zero, but the ordinary expectation is not defined.
- (d) The second moment is finite exactly when $2 < a$. For those values the mean also exists, so the variance is finite. Therefore

$$\text{Var}(X) < \infty \iff a > 2.$$

For example, $a = 3/2$ gives a well-defined mean equal to zero but an infinite second moment and hence no finite variance.

- (e) For $M \geq 0$,

$$\begin{aligned} \mathbb{P}(|X| > M) &= a \int_M^{\infty} (1+x)^{-a-1} dx \\ &= (1+M)^{-a}. \end{aligned}$$

This tends to zero for every $a > 0$, as every probability distribution's tail probability must. A moment weights the tail by a growing factor such as $|x|^r$. The unweighted tail probability can vanish even when that weighted tail integral diverges, so there is no contradiction.

4. **Using Jensen's inequality.** You may use Jensen's inequality for both convex and concave functions. For a concave function, the direction of the usual convex inequality is reversed. Let W be a positive random variable for which all expectations used below exist.

- (a) Determine which is larger, $\mathbb{E}[\log W]$ or $\log \mathbb{E}[W]$. State when the inequality is strict, and give an intuitive interpretation.
- (b) Determine which is larger, $\mathbb{E}[1/W]$ or $1/\mathbb{E}[W]$.
- (c) Let $0 < s < r$ and suppose $\mathbb{E}[|X|^r] < \infty$. By applying Jensen to a suitable function of $|X|^r$, prove that

$$\mathbb{E}[|X|^s] \leq (\mathbb{E}[|X|^r])^{s/r}.$$

Explain what this proves about the existence of lower-order moments.

- (d) Apply Jensen to a square to prove that $\text{Var}(X) \geq 0$. Under the usual finite-second-moment condition, when does equality hold?

Solution.

- (a) The logarithm is concave on $(0, \infty)$, so concave Jensen gives

$$\mathbb{E}[\log W] \leq \log \mathbb{E}[W].$$

Because the logarithm is strictly concave, the inequality is strict when W is not equal to a single constant with probability one (under the stated moment conditions). Intuitively, the logarithm gives especially low outcomes a large effect on the average. Therefore, taking logs before averaging gives no more than taking the log of the arithmetic average.

- (b) The function $g(w) = 1/w$ is convex on $(0, \infty)$ because $g''(w) = 2/w^3 > 0$. Convex Jensen therefore gives

$$\mathbb{E}\left[\frac{1}{W}\right] \geq \frac{1}{\mathbb{E}[W]}.$$

It is strict unless W equals a single constant with probability one.

- (c) Set $Y = |X|^r \geq 0$ and define $\phi(y) = y^{s/r}$. Since $0 < s/r < 1$, ϕ is concave. Concave Jensen gives

$$\mathbb{E}[|X|^s] = \mathbb{E}[\phi(Y)] \leq \phi(\mathbb{E}[Y]) = (\mathbb{E}[|X|^r])^{s/r}.$$

The right-hand side is finite by assumption. Thus existence of an absolute moment of order r implies existence of every absolute moment of lower positive order s .

- (d) The function $g(x) = x^2$ is convex, so

$$(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2].$$

It follows that

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \geq 0.$$

The square is strictly convex. With a finite second moment, equality in Jensen holds exactly when X equals a single constant with probability one, which is also exactly the condition for zero variance.

5. **Regression matrices and error covariance.** For three observations, let z be the explanatory variable and y be the outcome variable, with

$$z = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix},$$

and include an intercept in the regression of y on z .

- (a) Write down the design matrix X and state the dimensions of y , X and β .
- (b) Calculate $X'X$ and $X'y$. Using the formula $\hat{\beta} = (X'X)^{-1}X'y$, calculate the OLS intercept and slope.
- (c) Calculate the residual vector $\hat{u} = y - X\hat{\beta}$ and verify $X'\hat{u} = 0$.
- (d) Suppose, conditional on X , the error covariance matrix is

$$\Omega = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix}.$$

Use the diagonal and off-diagonal entries of Ω to describe its pattern of heteroscedasticity and dependence. Then calculate

$$\text{Var}(\hat{\beta} | X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

- (e) If the errors instead had covariance I_3 , their contribution to $\text{Var}(\hat{\beta} | X)$ would be different. Would this change the OLS coefficient formula used in part (b), or only its sampling variance? Explain using the formulas above.

Solution.

- (a) With an intercept in the first column,

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Here y is 3×1 , X is 3×2 , and $\beta = (\beta_0, \beta_1)'$ is 2×1 .

- (b) Direct multiplication gives

$$X'X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad X'y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Since

$$(X'X)^{-1} = \frac{1}{6} \begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix},$$

we obtain

$$\hat{\beta} = \frac{1}{6} \begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}.$$

Thus the fitted intercept is $7/6$ and the fitted slope is $1/2$.

- (c) The fitted values and residuals are

$$X\hat{\beta} = \begin{pmatrix} 7/6 \\ 5/3 \\ 13/6 \end{pmatrix}, \quad \hat{u} = \begin{pmatrix} -1/6 \\ 1/3 \\ -1/6 \end{pmatrix}.$$

Therefore

$$X'\hat{u} = \begin{pmatrix} -1/6 + 1/3 - 1/6 \\ 0(-1/6) + 1(1/3) + 2(-1/6) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

- (d) The diagonal variances are 1, 4 and 9, so the errors are heteroscedastic. All off-diagonal covariances are zero, so there is no cross-observation correlation encoded by this matrix. First,

$$X'\Omega X = \begin{pmatrix} 14 & 22 \\ 22 & 40 \end{pmatrix}.$$

Writing $A = (X'X)^{-1}$, the covariance calculation is

$$\begin{aligned} A(X'\Omega X)A &= \frac{1}{6} \begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix} \begin{pmatrix} 14 & 22 \\ 22 & 40 \end{pmatrix} \frac{1}{6} \begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 25/18 & -7/6 \\ -7/6 & 5/2 \end{pmatrix}. \end{aligned}$$

- (e) The OLS coefficient formula is still $(X'X)^{-1}X'y$. It depends on the observed X and y , so changing the assumed error covariance leaves the estimate for these data unchanged. The sampling covariance instead depends on Ω :

$$\text{Var}(\hat{\beta} \mid X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

Under $\Omega = I_3$, this reduces to $(X'X)^{-1}$; under the covariance in part (d), it gives the different matrix calculated there. Lecture 4 uses this distinction when choosing standard errors.